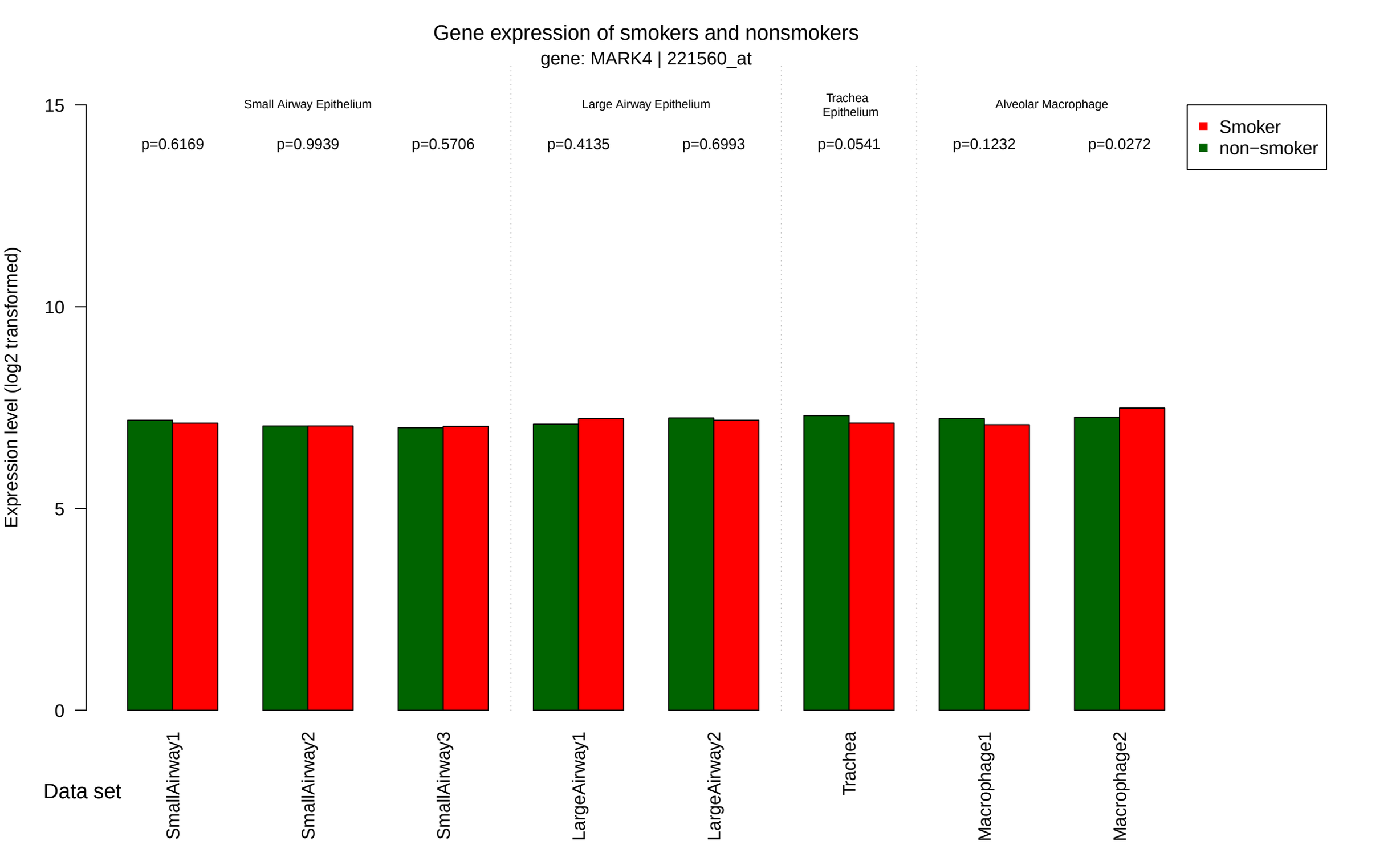
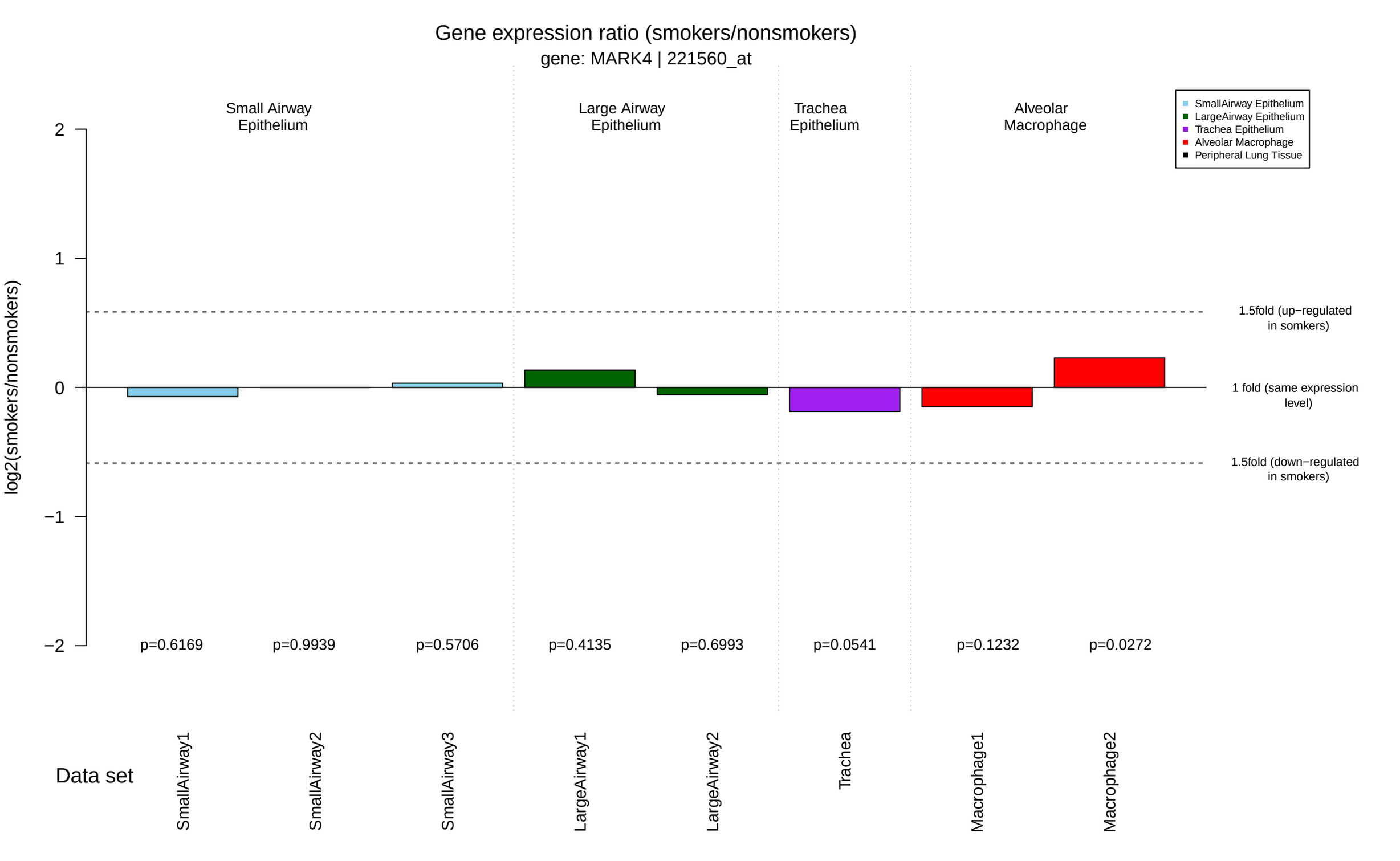
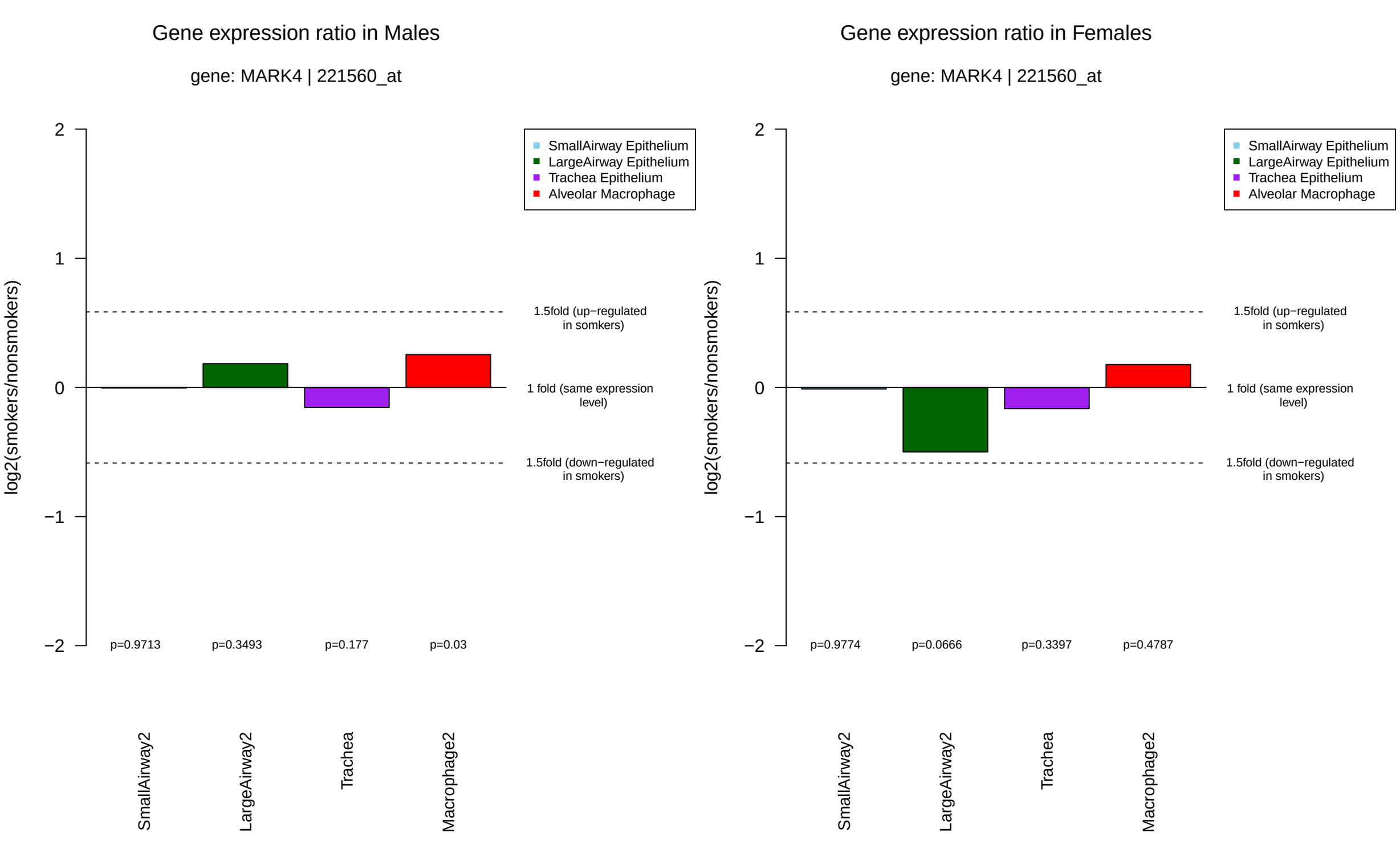
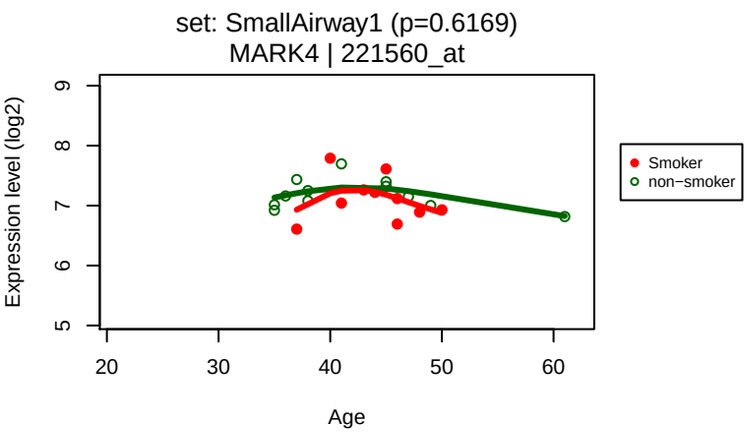
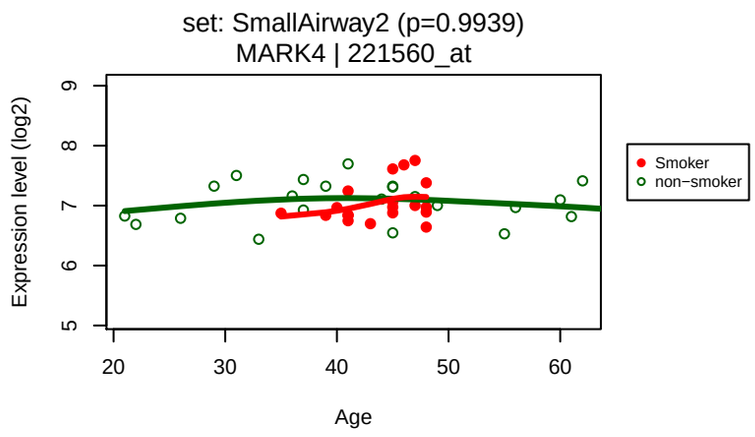
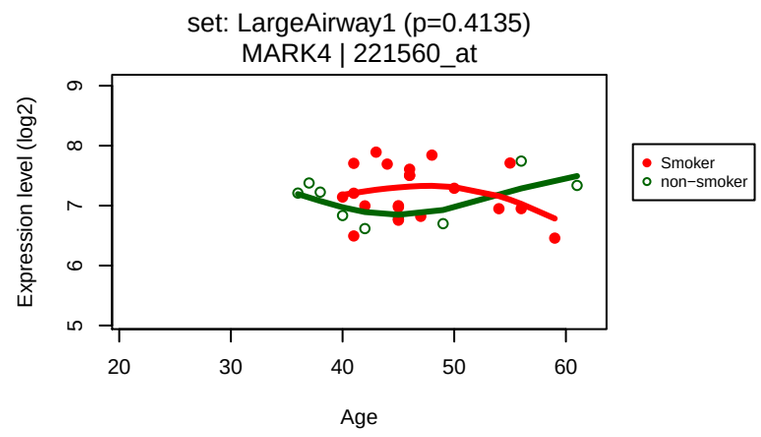

Gene expression of smokers and nonsmokers
gene: MARK4 | 221560_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6169
p=0.9939
p=0.5706
p=0.4135
p=0.6993
p=0.0541
p=0.1232
p=0.0272
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

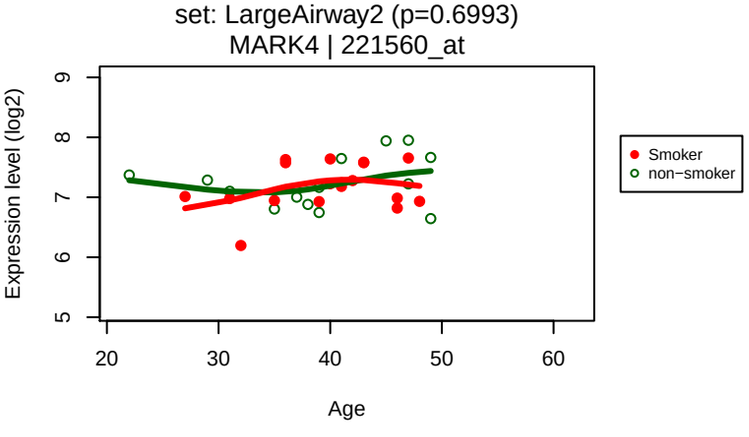

Gene expression ratio (smokers/nonsmokers)
gene: MARK4 | 221560_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6169
p=0.9939
p=0.5706
p=0.4135
p=0.6993
p=0.0541
p=0.1232
p=0.0272
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

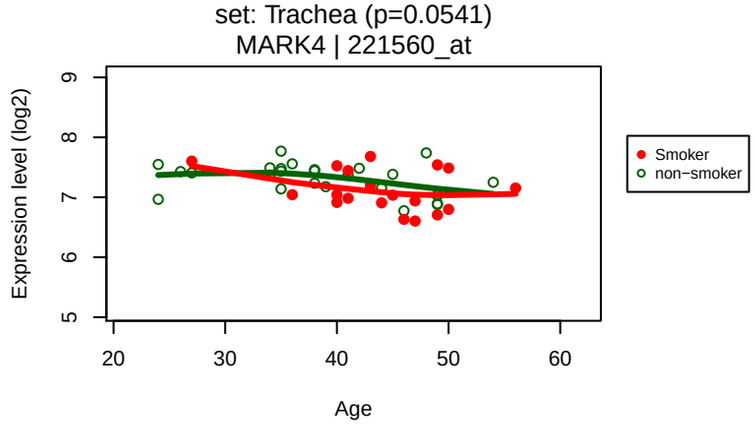

Gene expression ratio in Males
gene: MARK4 | 221560_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9713
p=0.3493
p=0.177
p=0.03
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MARK4 | 221560_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9774
p=0.0666
p=0.3397
p=0.4787
SmallAirway2
LargeAirway2
Trachea
Macrophage2

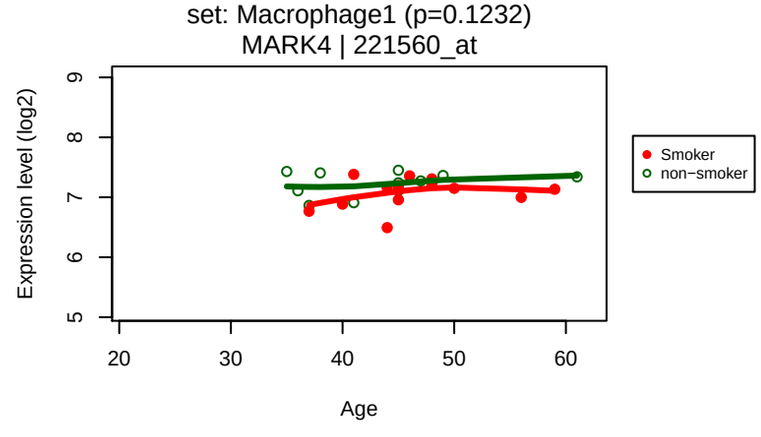

set: SmallAirway1 (p=0.6169)
MARK4 | 221560_at
Expression level (log2)
Age
Smoker
non−smoker

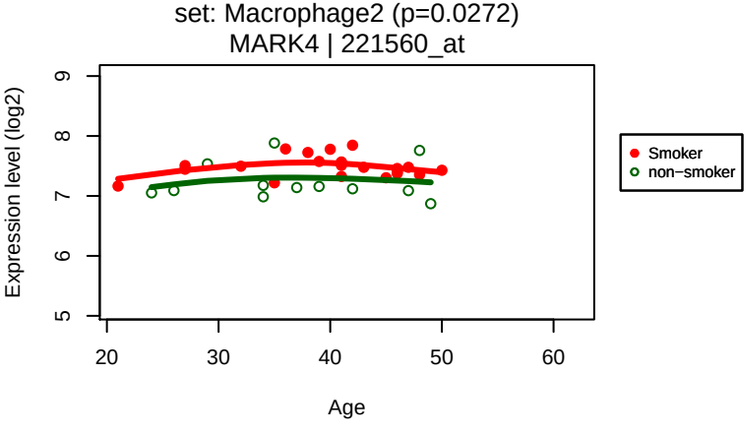

set: SmallAirway2 (p=0.9939)
MARK4 | 221560_at
Expression level (log2)
Age
Smoker
non−smoker

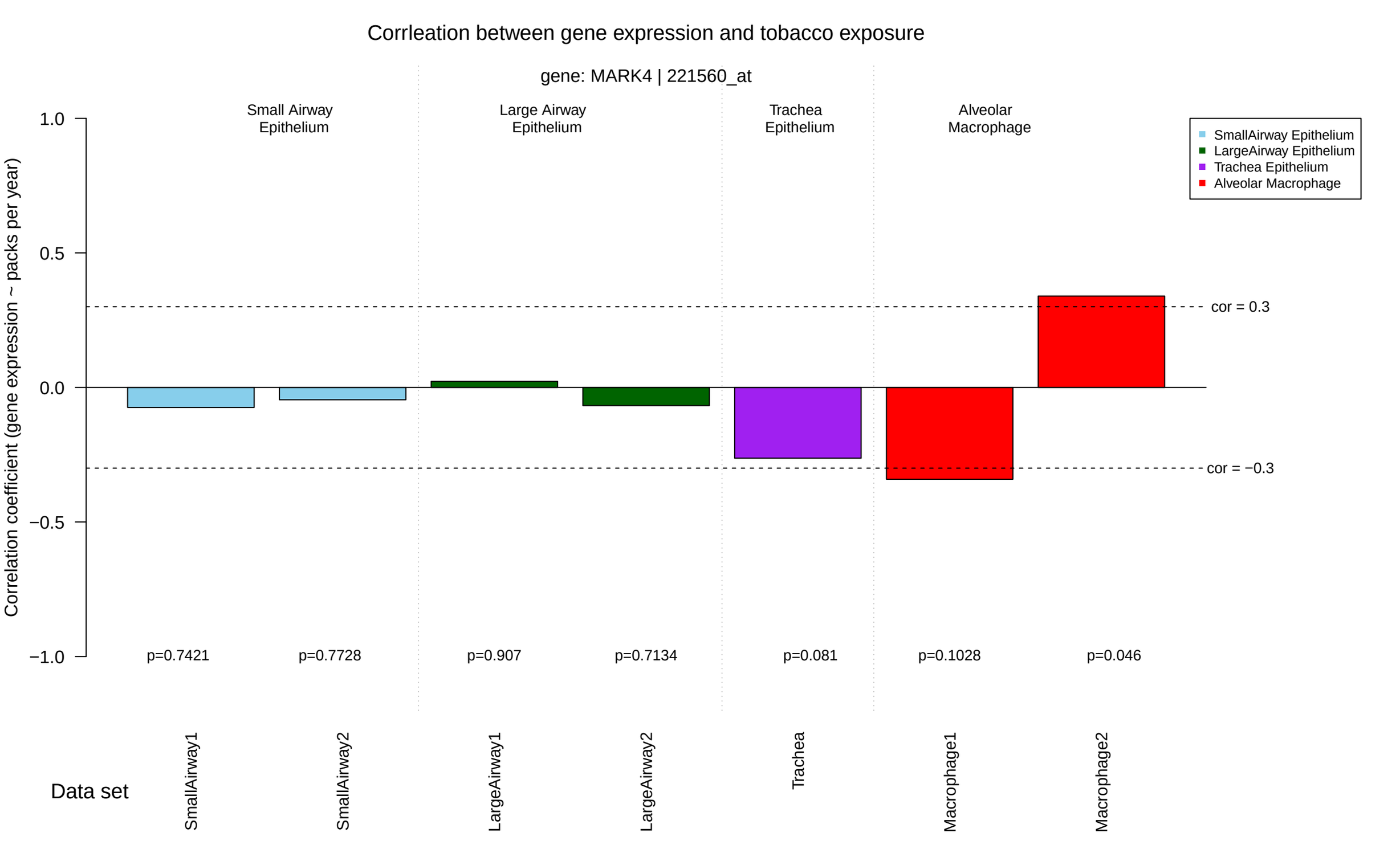

set: LargeAirway1 (p=0.4135)
MARK4 | 221560_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6993)
MARK4 | 221560_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0541)
MARK4 | 221560_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1232)
MARK4 | 221560_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0272)
MARK4 | 221560_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MARK4 | 221560_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7421
p=0.7728
p=0.907
p=0.7134
p=0.081
p=0.1028
p=0.046
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2